

SAYAN DUTTA

sayan_ua2504cdh152@iitp.ac.in | +91 8653128442 | Patna, Bihar

Github: <https://github.com/SAYANDUTTA8442> | LinkedIn: <https://linkedin.com/in/sayandutta8653128442>

BS-MS student in Computer Science and Data Analytics at IIT Patna with research experience in Large Language Models, empathetic dialogue generation, structured reasoning, and AI systems. Research Intern at BITS Pilani and co-author of an ACL ARR 2026 submission. Currently developing Blix, a privacy-first cognitive AI architecture for local AI agents. Research interests include affective computing, emotion-aware NLP, multimodal AI, reasoning systems, and responsible generative AI.

EDUCATION

Indian Institute of Technology Patna

2025 – 2030 (Expected)

BS-MS, Computer Science and Data Analytics (Integrated 5-Year)

- **CPI: 8.64 / 10.0**
- Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Optimization Fundamentals, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks

RESEARCH INTERESTS

Affective Computing, Emotion-aware NLP, Multimodal AI, Large Language Models, Reasoning Systems, Memory-Augmented AI, AI Agents, Continual Learning, Responsible AI, Privacy-Preserving AI

EXPERIENCE

Research Intern — BITS Pilani (Remote)

Apr 2026 – May 2026

Project: ECOT-ERG: Empathetic Dialogue Generation with Structured Chain-of-Thought | Supervisor: Dr. Srishti Gupta, PostDoctoral Fellow, Dept. of Computer Science

- Designed and implemented ECOT-ERG, a modular reasoning framework integrating structured chain-of-thought supervision, emotion prediction, cause identification, and T5-based response generation for empathetic dialogue.
- Designed modular emotion prediction and cause identification components enabling independent ablation of individual reasoning stages without full system retraining.
- Fine-tuned T5 models using FP16 mixed-precision training and constructed DPO preference pairs from ranked responses for empathy alignment.
- Conducted controlled ablation experiments evaluating structured reasoning, emotion prediction, cause identification, and DPO alignment using BERTScore, EASE, Cause-F1, and Relevance-Accuracy.
- Performed qualitative error analysis, LLM prompting experiments, and chain-based inference to evaluate model behavior.

PUBLICATIONS

Towards Causally Faithful Empathetic Dialogue Generation via Structured Appraisal Reasoning

Srishti Gupta, Ayushman Sarkar, Sayan Dutta, Shashank Vishwakarma, Rishikesh Raj, Arghya Jana, Nagandla Tarun Teja
ACL ARR 2026 (Under Review) : [OpenReview: https://openreview.net/forum?id=nq94vrulUm](https://openreview.net/forum?id=nq94vrulUm)

Contributions: Developed and trained the ECOT-ERG architecture integrating structured chain-of-thought reasoning, emotion prediction, and cause identification; constructed reasoning-chain supervision targets; conducted controlled ablation experiments; and performed qualitative and quantitative evaluation.

PROJECTS

Blix – Privacy-First Cognitive AI Architecture

2024 – Present

Independent Project | Python, PyTorch, HuggingFace Transformers, sqlite-vec, FastAPI, Pydantic

Link: <https://github.com/SAYANDUTTA8442/blix.ai>

- Currently developing Blix, a privacy-first cognitive AI architecture designed for fully local operation on consumer hardware, integrating Hybrid Graph-based Semantic Hierarchical Memory (HGSHM), causal reasoning, truth maintenance, and reflective metacognition for memory-augmented AI agents - validated by 1,962 passing tests.
- Built Adaptive Dual Memory Architecture (ADMA), a gradient-free adaptation layer using Thompson-sampling contextual bandits to tune retrieval weighting, planning aggressiveness, and prompt style from live reward signals, with automatic rollback on confidence regression and no retraining loop required.
- Implemented a beam-search planner with global-workspace-style coordination (verifier, evaluator, specialist consensus voting) and a failure-memory/replanning loop for autonomous recovery from failed tasks.
- Developed a curiosity-driven research loop (knowledge-gap detection → hypothesis generation → experiment planning) enabling self-directed exploration without external supervision.
- Exposed the system via a FastAPI layer with 20 routers for programmatic access to memory, planning, causal reasoning, and reflection subsystems.

- Built a custom evaluation harness (StateBench) covering state tracking, contradiction detection, retrieval fidelity, replan success rate, and confidence calibration, enabling independent subsystem validation rather than relying solely on end-to-end testing.
- Designed a dependency-injected ablation framework (ablation_v3) allowing any of 7 ADMA components to be swapped or stubbed independently, enabling controlled experiments to isolate which architectural choices drive performance gains.
- Implemented an agent execution loop (executor, task runtime, tool-reliability tracking, tool-success prediction) alongside a document/media ingestion pipeline (PDF, DOCX, OCR) for grounding agent knowledge in external sources.

ASES - AI-Driven Precision Agriculture Advisory Platform (Capstone Project)

Jan 2026 – May 2026

Python, Scikit-learn, Google Gemini API, Streamlit, SQLite

Live Link: <https://sayanduttaiitpatnacda107.streamlit.app/>

- Developed crop and fertilizer recommendation modules using feature-normalized KNN models over multidimensional soil and environmental attributes.
- Enhanced recommendation quality by integrating LLM-based contextual advisory generation through structured prompting with Google Gemini API.
- Designed a lightweight end-to-end application with Streamlit frontend and SQLite backend for persistent storage and local deployment.
- Established a modular pipeline separating prediction, recommendation, and natural-language advisory components for extensible agricultural decision support.

TECHNICAL SKILLS

Languages: Python, C++, C, Java

ML / DL: PyTorch, Hugging Face Transformers, Scikit-learn, Seq2Seq Training

Generative AI: Large Language Models (LLMs), Chain-of-Thought (CoT), Direct Preference Optimization (DPO), Retrieval-Augmented Generation (RAG), Prompt Engineering

Responsible AI: Privacy-Preserving AI, Responsible AI, Local AI Inference

NLP: Sentence Transformers, T5, Empathetic Dialogue Systems, BERTScore, EASE, Cause-F1

AI Systems: FAISS, FastAPI, REST APIs, Pydantic, sqlite-vec

Tools & Platforms: Git, Linux, CUDA, FP16 Mixed Precision, Jupyter Notebook, Docker, Google Cloud

Databases: MySQL, SQLite

CERTIFICATIONS

Google Cloud Skills Boost — 70 completions (Apr–Jun 2025), including Introduction to Large Language Models, Generative AI, and Responsible AI; Gemini-based GenAI application development; MLOps for AI agents.

Full profile: https://www.skills.google/public_profiles/7134a42a-d7bf-453e-ab38-0570038bf2a8

LEADERSHIP

Lead Google Student Ambassador

2025 – Present

Indian Institute of Technology Patna

- Coordinated technical outreach initiatives serving 100+ students through AI/ML workshops and developer events.
- Led the IIT Patna Google Student Ambassador chapter, coordinating technical events, workshops, and student-facing initiatives.
- *Chapter Website:* <https://gsaiitp.netlify.app>

ACHIEVEMENTS

- Co-author of an ACL ARR 2026 submission on structured chain-of-thought reasoning for empathetic dialogue generation.
- Selected as a Research Intern under Dr. Srishti Gupta, Department of Computer Science, BITS Pilani.
- **Lead Google Student Ambassador at IIT Patna**, leading technical outreach initiatives reaching 100+ students.
- **Developing Blix**, an independent privacy-first cognitive AI architecture for fully local AI operation on consumer hardware.